AI and the Evolution of Software Industry Roles

Introduction Report

Forecasting Emerging and Declining Roles in Software & IT (2025-2040)

This report presents a structured analysis of the roles within the global Software and Information Technology sector that are projected to **emerge**, **evolve**, or **decline** between **2025 and 2040** under the accelerating influence of artificial intelligence (AI), automation, multi-agent systems, and advanced machine learning models.

Our analysis synthesizes:

- Observations from international labor market datasets
- Research from institutions such as **World Economic Forum (WEF)**, **McKinsey**, **Gartner**, and **Google Research**
- AI-driven task automation trendlines
- Corporate digital transformation reports spanning the U.S., Europe, and East Asia (particularly China)

Across all regions examined, AI appears to be reshaping job families rather than eliminating entire fields. Entry-level and routine operational tasks face the highest risk of automation, while roles involving **AI orchestration**, **governance**, **security**, **explainability**, and **systems engineering** demonstrate the strongest growth potential.

The period **2025-2040** is characterized by a shift from human-performed routine work toward hybrid ecosystems where humans supervise, validate, and architect intelligent systems. The emerging roles require interdisciplinary skill sets involving software engineering, machine learning, ethics, compliance, human AI interaction, and cloud-native infrastructure.

WARNING !

Scientific Validity and Forecasting Disclaimer

This analysis **does not claim absolute predictive certainty**.
It is based on current machine learning capabilities, observed industry automation patterns, and extrapolated technological adoption curves.

Because AI development is nonlinear and influenced by geopolitical, economic, and regulatory variables:

The 15 year forecast may shorten or extend depending on future breakthroughs or slowdowns in AI capability growth.

This report should therefore be interpreted as a **high-confidence directional forecast**, not a deterministic prediction.

Informational Text About the Table

- The Roles indicated by **"Title"** in the table represent the software and IT professionals employed by current BigTech and indie Startups.

- **“Status”** is divided into three categories. **"Declining"** indicates the possibility or expected decrease in the situation. **"Transforming"** indicates the changing status of existing roles in the sector. **"Emerging"** indicates the expected roles to emerge in the sector.

- The **"Estimited Timeline"** covers the estimated time period from 2025 to 2040.

- **“Scientific/Sectoral Rationale”** gives the explanation of why it is evolving towards a Declining, Transforming and Emerging direction.

	Title	Status	Estimated Timeline	Scientific / Sectoral Rationale
1	Junior Software Developer	Declining	2025–2030	LLM-powered code generation, agentic automation, and code-as-a-service reduce the demand for routine, entry-level coding tasks (Sheet observation + automation literature).
2	Computer Programmer (Routine Coding)	Declining	2025–2030	Enterprise routine coding is replaced by automated refactoring and code-generation agents; human work shifts toward supervision and integration.
3	Manual QA / Test Specialist	Declining	2025–2028	Autonomous test agents and LLM-driven self-healing test frameworks absorb repetitive testing tasks; QA shifts to strategy and validation.
4	Test Case Writer (manual)	Declining	2025–2028	LLMs generate standard test cases; humans focus on edge cases, risk-based scenarios, and validation.
5	Helpdesk / IT Support (L1)	Declining	2025–2029	Self-service automation, chatbots, and AI troubleshooting flows resolve first-level support issues.
6	Technical Support Call Operator	Declining	2025–2029	LLM-driven call agents and voice assistants handle repetitive support interactions.
7	Data Entry Operator (IT)	Declining	2025–2028	RPA + OCR + automated data-cleaning pipelines eliminate manual entry tasks.
8	Web Designer (Template-based)	Declining	2025–2030	No-code AI website builders automate template-driven web design; human designers move toward strategy and creativity.
9	On-prem System Administrator (legacy)	Transforming	2026–2035	Cloud, IaC, CloudOps, and autonomous infrastructure shift the role into cloud-native engineering.
10	Network Administrator (routine)	Transforming	2026–2034	SDN, automated configuration, and AI-based network optimization reduce routine network management tasks.

11	DBA (routine ops)	Transforming	2026–2034	Self-healing cloud DBs and serverless architectures automate basic DBA work; humans focus on architecture and optimization.
12	Desktop Hardware Technician	Declining	2025–2032	Modular hardware, remote diagnostics, and zero-touch provisioning reduce field technician needs.
13	Data Center Operator (NOC L1)	Declining	2025–2029	AIOps platforms automate monitoring, alerting, and first-level resolution in NOC environments.
14	Routine Systems Analyst (waterfall)	Transforming	2026–2033	AI-supported requirements engineering moves the role toward data-informed product analysis.
15	Reporting Specialist (basic BI)	Declining	2025–2030	Auto-insight BI systems automatically generate reports; analysts shift toward strategic interpretation.
16	SEO Content Operator (manual)	Declining	2025–2028	Generative AI and automated ranking optimization reduce manual SEO content workflows.
17	Social Media Content Operator	Declining	2025–2028	AI-driven content orchestration automates publishing, basic design, and scheduling tasks.
18	Technical Writer (routine docs)	Declining	2025–2029	AI doc generators automate routine documentation; human writers focus on XAI and high-risk system descriptions.
19	Onsite Setup Technician (PC)	Declining	2025–2029	Zero-touch deployment and automated provisioning reduce on-site installation work.
20	IT Inventory Manager (manual)	Declining	2025–2029	RFID automation and AI-based asset tracking replace manual inventory operations.
21	QA UAT Coordinator (manual)	Declining	2025–2029	AI-powered UAT orchestration reduces the need for manual coordination.
22	Data Labeler (manual)	Transforming	2025–2032	Auto-tagging and synthetic datasets reduce manual labeling, but governance and quality-review roles expand.

23	Technical Translator (IT)	Declining	2025–2029	Real-time LLM translation reduces the need for manual technical translation; cultural quality assurance roles rise instead.
24	Basic Script Developer (RPA / Macro)	Declining	2025–2028	Agentic RPA + LLM agents generate basic automation scripts automatically.
25	AI Agent Orchestrator	Emerging	2025–2035	Needed to manage multi-agent AI architectures, task allocation, verification, and inter-agent communication.
26	Generative AI Security Expert	Emerging	2025–2032	Required to defend against model poisoning, prompt injection, and adversarial ML threats.
27	LLMOps Engineer	Emerging	2025–2030	LLM deployment, optimization, quantization, telemetry, and drift-mitigation require specialization.
28	Observability Architect (AI systems)	Emerging	2026–2034	AI-era systems require unified metric-log-trace correlation and model telemetry observability.
29	AI-Augmented QA Strategist	Emerging	2025–2031	Humans must validate AI-generated tests, assess coverage gaps, and manage production-grade quality strategies.
30	Platform Engineering Specialist (IDP)	Emerging	2025–2032	Organizations need scalable Internal Developer Platforms for self-service DevOps workflows.
31	Semantic Search Developer	Emerging	2025–2031	Vector DBs, embeddings, and RAG architectures require specialized search engineers.
32	AI Product Manager / Strategist	Emerging	2025–2030	Needed to connect AI capabilities with business strategy, ROI, and regulatory risk.
33	AI Compliance & Ethics Analyst	Emerging	2027–2035	Driven by the EU AI Act and global regulatory frameworks requiring ethical oversight.

34	Autonomous Infrastructure Specialist / CloudOps Architect	Emerging	2026–2035	Self-healing infrastructure, FinOps, and autonomous cloud governance demand specialized architects.
35	AI Chatbot Trainer / Knowledge Engineer	Emerging	2025–2029	Conversational agents require knowledge curation, domain-tuning, and safety scaffolding.
36	Model Performance & Integrity Analyst	Emerging	2025–2032	Needed to monitor model drift, data shift, inference reliability, and bias across production systems.
37	Explainability & Transparency Designer	Emerging	2025–2032	Growing global requirements for interpretable AI systems drive XAI-focused design roles.
38	Prompt Engineer / Prompt Architect	Emerging (short-term)	2025–2028	Needed in the short term to guide and optimize LLM behavior; role may partially decline as models self-optimize.
39	AI Security Specialist (Model & Data Protection)	Emerging	2025–2031	Model protection, adversarial defense, and data privacy specialization increase in demand.
40	Digital Twin Engineer	Emerging	2026–2034	AI + IoT integration creates demand for simulation and industrial twin modeling specialists.
41	Human-AI Interaction Designer	Emerging	2025–2032	Multimodal UX and AI-agent interaction models require new human-centered design roles.
42	AI Sustainability Analyst	Emerging	2026–2034	Green-AI efficiency, carbon-cost optimization, and energy modeling drive demand for this specialty.
43	AI-enabled Cybersecurity Data Scientist	Emerging	2025–2032	ML-based threat detection and autonomous cyber defense require hybrid data science + security roles.

What Does This Table Mean ?

The Future of the Software and IT Sector: A Great Transformation (2025-2040)

Under the accelerating influence of artificial intelligence (AI), automation, and multi-agent systems, the global Software and Information Technology (IT) sector is set to undergo a comprehensive change between 2025 and 2040. This period is characterized by a shift from human performed routine work to a **hybrid operating model** where humans **architect, supervise, and ethically govern intelligent systems.**

Automation of Routine Tasks and Declining Roles (2025-2030)

In the first phase of this transformation, **entry level and routine operational tasks** will face the highest risk of automation. Roles like **Junior Software Developer**, **Computer Programmer (Routine Coding)**, **Manual QA Test Specialist**, and **L1 Helpdesk Support** will rapidly decrease between 2025 and 2030 due to LLM-powered code generation and autonomous testing agents.

Roles Transforming into Strategic Areas (2026-2035)

Traditional roles will be forced to **transform** under the influence of AI. For instance, **On prem System Administrators** and **Routine Database Administrators (DBAs)** will give way to more architectural and strategic areas such as **CloudOps Architecture**, **Infrastructure as Code (IaC)** expertise, and **FinOps** (Financial Optimization). The fundamental value in these roles will shift from technical execution to **design, automation, and system optimization** skills.

New and Critical Expertise Areas (2025-2040)

Managing the complexity of AI systems will create entirely new and high-demand roles. These roles are primarily concentrated around **AI orchestration, security, and ethical governance**:

1. **AI Agent Orchestrator:** This will be one of the most critical roles, managing systems composed of multiple autonomous AI agents, delegating tasks, and verifying agent steps towards a goal.
2. **LLM Operations Engineer (LLMOps):** Will be responsible for scaling, monitoring performance, optimizing costs, and controlling model drift for LLMs in production environments.
3. **Generative AI Security Specialist:** Will develop defensive mechanisms against AI-specific cyber threats such as prompt injection and model poisoning.
4. **Platform Engineering Specialist:** Responsible for building **Internal Developer Platforms (IDP)** that boost developer efficiency and facilitate AI integration.
5. **AI Compliance and Ethics Analyst:** Driven by global regulations like the EU AI Act, this role will oversee the compliance of internal AI systems with legal and ethical standards.
6. **Human AI Interaction Designer:** Will design the interaction between users and Multimodal AI agents to be intuitive and collaborative.
7. **XAI Designer / Transparent AI Architect:** Focused on designing systems that can explain the rationale behind AI decisions, driven by growing transparency requirements.

The Enduring Value of the Human Role

As automation rises, the value of IT professionals is shifting from **technical application** to **strategic judgment, ethical oversight, and human resilience**. The most resilient and lasting roles will require uniquely human capabilities such as **Emotional Intelligence**, **Empathy**, and **Perspective Taking**, while also demanding **strategic leadership** in highstakes areas like **ML&based threat detection in cybersecurity**. In short, the sector is moving from **code execution to AI orchestration and governance.**

Author Opinions

In short, just like the Web revolution, this revolution will continue to evolve and take different forms. As software developers, we must adapt to this transformation as long as sufficient energy resources are provided and maintained sustainably. Don't be misled by vibecoding style online rhetoric saying, “Don’t do this! It will take your job!” In reality, there is no reason why someone who uses AI more effectively shouldn't come out ahead.

As with every technological advancement, this one will inevitably cause some harm. In fact, the future might bring even more harm than progress. However, I don’t want to worry you at this moment. Still, the most fundamental law of biology always remains true: Survival of the fittest.